undercuts Laplacian determinism because we fundamentally cannot know the precise positions and velocities of the constituents of the universe. Instead, these classical properties are replaced by quantum wave functions, which tell us only the probability that any given particle is here or there, or that it has this or that velocity.

The downfall of Laplace's vision, however, does not leave the concept of determinism in total ruins. Wave functions—the probability waves of quantum mechanics—evolve in time according to precise mathematical rules, such as the Schrödinger equation (or its more precise relativistic counterparts, such as the Dirac equation and the Klein-Gordon equation). This informs us that quantum determinism replaces Laplace's classical determinism: Knowledge of the wave functions of all of the fundamental ingredients of the universe at some moment in time allows a "vast enough" intelligence to determine the wave functions at any prior or future time. Quantum determinism tells us that the probability that any particular event will occur at some chosen time in the future is fully determined by knowledge of the wave functions at any prior time. The probabilistic aspect of quantum mechanics significantly softens Laplacian determinism by shifting inevitability from outcomes to outcome-likelihoods, but the latter are fully determined within the conventional framework of quantum theory.

In 1976, Hawking declared that even this softer form of determinism is violated by the presence of black holes. Once again, the calculations behind this declaration are formidable, but the essential idea is fairly straightforward. When anything falls into a black hole, its wave function gets sucked in as well. But this means that in the quest to work out wave functions at all future times, our "vast enough" intelligence will be irreparably shortchanged. To predict the future fully we need to know all wave functions fully today. But, if some have escaped down the abyss of black holes, the information they contain is lost.

At first sight, this complication arising from black holes may not seem worth worrying about. Since everything behind the event horizon of a black hole is cut off from the rest of the universe, can't we just completely ignore anything that is unfortunate enough to have fallen in? Philosophically, moreover, can't we tell ourselves that the universe